TERRASENTINEL AI — GEOSPATIAL EVIDENCE DOSSIER

INVESTIGATIVE REPORT | CASE TS-BAL-2026-001 | ISSUED: 2026-10-20 16:18:18 UTC

1. TARGET JURISDICTION & CASE IDENTIFICATION

Case Identifier:	TS-BAL-2026-001	**Candidate Category:**	Candidate land disturbance consistent with
Target District:	Ballari (Bellary) District	**Jurisdiction Source:**	Survey of India / Karnataka Mines & Geolo
Centroid Coords:	15.17720° N, 76.94200° E	**Footprint Relation:**	UNINVENTORIED_CANDIDATE
Estimated Disturbance Area:	0.64 ha (6,400 m²)	**Calculation Method:**	Geodesic ellipsoidal polygon quadrature (V
Permit / Lease Status:	Unknown (State Cadastre unlinked)	**Review Priority Score:**	95 / 100

2. SATELLITE OBSERVATION METADATA (SENTINEL-2 L2A)

PARAMETER	OBSERVATION T1 (PRE-DISTURBANCE)	OBSERVATION T2 (POST-DISTURBANCE)
Acquisition Date (UTC):	2023-02-25 05:34:46	2024-02-25 05:34:48
Sentinel-2 Scene ID:	S2A_43PFS_20230225_0_L2A...	S2B_43PFS_20240225_0_L2A...
Platform & Tile:	Sentinel-2A | Tile 43PFS	Sentinel-2B | Tile 43PFS
Cloud Cover / Sun Angle:	1.2% | 54.8°	0.06% | 56.1°

3. SPECTRAL SHIFT & MULTI-CRITERIA EVIDENCE SCORES

- Delta NDVI (Vegetation Loss): -1.197 (Pre: 0.52 -> Post: 0.16)
- Delta BSI (Bare Soil / Mineral Rise): 0.865 (Pre: -0.09 -> Post: 0.31)
- Delta NDMI (Moisture Depletion): -0.22
- Spatial Coherence Evidence Score: 80/100 (Compact contiguous excavation pit)
- Observation Atmospheric Quality: 94/100 (SCL cloudless pass)
- Environmental Buffer Proximity: Within 400m statutory buffer of Narihalla Catchment Buffer Zone

4. FALSE-POSITIVE EVALUATION & ALTERNATIVE HYPOTHESES

- Agricultural land clearing / pre-sowing soil exposure.
- Civil works, road corridor excavation, or construction ground leveling.
- Seasonal vegetative drying or low-intensity scrub fire scar.

5. HUMAN REVIEW STATUS & FIELD VERIFICATION AUDIT

Current Review Status: UNREVIEWED Reviewed by: Unassigned | Date: Pending review

Mandatory Ground Audit Checklist for Mining Inspectorate:

[ ] 1. Verify existence of demarcated boundary survey boundary pillars (boundary stones/pegs).
[ ] 2. Inspect whether heavy earth-moving machinery (excavators, loaders, crushers) is present inside the detected polygon.
[ ] 3. Check official mineral dispatch registers and electronic transit passes (e-Permits).
[ ] 4. Measure depth of extraction benching and highwall geometry using handheld laser distometer.
[ ] 5. Collect GPS ground truth coordinates at northernmost and southernmost pit perimeters.

STATUTORY GEOSPATIAL INTELLIGENCE NOTICE & LIMITATION OF LIABILITY:

This dossier represents candidate land-disturbance signals extracted from Sentinel-2 satellite reflectance data. Remote sensing cannot independently establish title ownership, mineral concessions, statutory permits, or legal guilt. It is generated strictly as an objective technical aid for inspection prioritization under district administrative authority.